EXERCISE

Flow Matching and VLA

Which clock is running?

k: physical robot step; $\tau \in [0, 1]$: generator time. One action chunk is refined while the camera observation stays fixed.

A. Every flow step moves the robot.

B. Flow steps refine a candidate chunk.

C. Flow time is the camera frame index.

π_0, §IV and Appendix A-D

SOLUTION

Flow Matching and VLA

Which clock is running?

Solution 1 / 1

Separate computation from execution

k: robot time; τ: generation time; A: candidate action chunk.

1. The generator changes the candidate A while holding the observation at robot step k fixed.
2. The robot executes selected entries only after generation.
3. B is correct. A confuses numerical refinement with execution; C identifies two different clocks.

π_0, §IV and Appendix A-D

RESULT

Flow Matching and VLA

Which clock is running?

k: physical robot step; $\tau \in [0, 1]$: generator time. One action chunk is refined while the camera observation stays fixed.

A. Every flow step moves the robot.

B. Flow steps refine a candidate chunk. ✓

C. Flow time is the camera frame index.

π_0, §IV and Appendix A-D

EXERCISE

Probability Flows

Transfer: a different velocity

The initial law is $X_0 \sim \mathcal{N}(1, 4)$ (variance 4). Each state x_τ has velocity $dx_\tau/d\tau = -3$. Find the law at generation time $\tau = 1$.

A Gaussian mean -2, variance 4

B Gaussian mean -2, variance 16

C Gaussian mean 4, variance 4

Lipman et al., §2, Eqs. (1)–(4)

SOLUTION

Probability Flows

Transfer: a different velocity

Integrate before transforming the law

X_0: initial random value; X_1: final value.

1. $X_1 = X_0 + \int_0^1 (-3)\, d\tau = X_0 - 3.$
2. $\mathbb{E}[X_1] = 1 - 3 = -2.$
3. $\mathrm{Var}(X_1) = \mathrm{Var}(X_0) = 4.$

Lipman et al., §2, Eqs. (1)–(4)

SOLUTION

Probability Flows

Transfer: a different velocity

Solution 2 / 2

Choose A

$\mathcal{N}(\mu, \sigma^2)$ uses variance as its second argument.

1. A follows because a constant shift leaves centered differences unchanged.
2. B incorrectly scales the spread. C reverses the velocity sign.

Lipman et al., §2, Eqs. (1)–(4)

RESULT

Probability Flows

Transfer: a different velocity

The initial law is $X_0 \sim \mathcal{N}(1, 4)$ (variance 4). Each state x_τ has velocity $dx_\tau/d\tau = -3$. Find the law at generation time $\tau = 1$.

- **A** Gaussian mean -2, variance 4 ✓
- **B** Gaussian mean -2, variance 16
- **C** Gaussian mean 4, variance 4

Lipman et al., §2, Eqs. (1)–(4)

EXERCISE

Probability Flows

Does constant velocity freeze density?

A smooth starting density p_0 translates at rate $v = 2$: $p_\tau(x) = p_0(x - 2\tau)$. At a fixed position x, does density stay constant as τ changes?

A. Yes, because velocity is constant.

B. No, because the density profile moves.

C. No, because total probability grows.

Lipman et al., Appendix B, Eq. (26)

SOLUTION

Probability Flows

Does constant velocity freeze density?

Differentiate the translated profile

p_0 is differentiable; prime means derivative of its scalar argument.

1. $\partial_\tau p_0(x - 2\tau) = -2p_0'(x - 2\tau)$.
2. $-\partial_x(2p_\tau) = -2p_0'(x - 2\tau)$.
3. These sides match, so translation satisfies continuity.

Lipman et al., Appendix B, Eq. (26)

SOLUTION

Probability Flows

Does constant velocity freeze density?

Solution 2 / 2

Choose B

x stays fixed while the profile passes it.

---

1. B: the density at x can change even with constant velocity.
2. A confuses constant velocity with constant density. C violates conserved total mass.

Lipman et al., Appendix B, Eq. (26)

RESULT

Probability Flows

Does constant velocity freeze density?

A smooth starting density p_0 translates at rate $v = 2$: $p_\tau(x) = p_0(x - 2\tau)$. At a fixed position x, does density stay constant as τ changes?

A. Yes, because velocity is constant.

B. No, because the density profile moves. ✓

C. No, because total probability grows.

Lipman et al., Appendix B, Eq. (26)

EXERCISE

Probability Flows

Transfer: contracting toward a target mean

Initial $X_0 \sim \mathcal{N}(0, 4)$ has variance 4. The state at generation time τ is $X_\tau = 3\tau + (1 - \tau/2)X_0$. Find mean and variance at $\tau = 1$.

A. Mean 3, variance 1

B. Mean 3, variance 2

C. Mean 0, variance 1

Lipman et al., §2; Goodfellow et al., §3.8

SOLUTION

Probability Flows

Transfer: contracting toward a target mean

Apply affine moment rules

$X_1 = 3 + X_0/2$; initial variance is 4.

1. $\mathbb{E}[X_1] = 3 + (1/2)0 = 3.$
2. $\mathrm{Var}(X_1) = (1/2)^2(4) = 1.$
3. The endpoint scale 1/2 stays positive, so this affine map remains invertible.

Lipman et al., §2; Goodfellow et al., §3.8

SOLUTION

Probability Flows

Transfer: contracting toward a target mean

Solution 2 / 2

Choose A

$\mathcal{N}(3, 1)$ is the final Gaussian.

1. A squares the scale when transforming variance.
2. B multiplies variance by the scale only. C ignores translation.

Lipman et al., §2; Goodfellow et al., §3.8

RESULT

Probability Flows

Transfer: contracting toward a target mean

Initial $X_0 \sim \mathcal{N}(0, 4)$ has variance 4. The state at generation time τ is $X_\tau = 3\tau + (1 - \tau/2)X_0$. Find mean and variance at $\tau = 1$.

- **A** Mean 3, variance 1 ✓
- **B** Mean 3, variance 2
- **C** Mean 0, variance 1

Lipman et al., §2; Goodfellow et al., §3.8

EXERCISE

Conditional Paths

Transfer: another pair

Noise $x_0 = (2, -1)^T$, data $x_1 = (-2, 3)^T$, generation time $\tau = 1/4$. Choose the intermediate state x_τ and target velocity u.

A ((1, 0), (−4, 4))

B ((−1, 2), (−4, 4))

C ((1, 0), (−1, 1))

Tong et al., §3.2.2

SOLUTION

Conditional Paths

Transfer: another pair

Solution 1 / 2

Compute the rate, then the state

$u = x_1 - x_0$; $x_\tau = x_0 + \tau u$.

---

1. $u = (-2 - 2, 3 - (-1))^T = (-4, 4)^T$.
2. $x_{1/4} = (2, -1)^T + (1/4)(-4, 4)^T = (1, 0)^T$.

Tong et al., §3.2.2

SOLUTION

Conditional Paths

Transfer: another pair

Solution 2 / 2

Choose A

$\tau u = (-1, 1)^T$ is an increment, not the rate.

---

1. A gives both quantities correctly.
2. B evaluates at τ=3/4. C replaces the rate by a quarter-step increment.

Tong et al., §3.2.2

RESULT

Conditional Paths

Transfer: another pair

Noise $x_0 = (2, -1)^T$, data $x_1 = (-2, 3)^T$, generation time $\tau = 1/4$. Choose the intermediate state x_τ and target velocity u.

A ((1, 0), (−4, 4)) ✓

B ((−1, 2), (−4, 4))

C ((1, 0), (−1, 1))

Tong et al., §3.2.2

EXERCISE

Conditional Paths

Is the noise coefficient a variance?

Noise $X_0 \sim \mathcal{N}(0, 1)$, fixed data $y = 4$. The state is $X_\tau = (1 - \tau)X_0 + \tau y$. Find its mean and variance at time $\tau = 1/2$.

A Mean 2, variance 1/2

B Mean 2, variance 1/4

C Mean 4, variance 1/4

Lipman et al., §4, Eqs. (10)–(14)

SOLUTION

Conditional Paths

Is the noise coefficient a variance?

Solution 1 / 2

Transform the standard Gaussian

$X_{1/2} = X_0/2 + 2.$

---

1. $\mathbb{E}[X_{1/2}] = 2.$
2. $\mathrm{Var}(X_{1/2}) = (1/2)^2\,\mathrm{Var}(X_0) = 1/4.$
3. The standard deviation is 1/2.

Lipman et al., §4, Eqs. (10)–(14)

SOLUTION

Conditional Paths

Is the noise coefficient a variance?

Solution 2 / 2

Choose B

$y = 4$ is fixed; it supplies no conditional variance.

---

1. B keeps the standard deviation and variance distinct.
2. A uses the noise coefficient as variance. C uses the final rather than intermediate mean.

Lipman et al., §4, Eqs. (10)–(14)

RESULT

Conditional Paths

Is the noise coefficient a variance?

Noise $X_0 \sim \mathcal{N}(0,1)$, fixed data $y = 4$. The state is $X_\tau = (1-\tau)X_0 + \tau y$. Find its mean and variance at time $\tau = 1/2$.

A Mean 2, variance 1/2

B Mean 2, variance 1/4 ✓

C Mean 4, variance 1/4

Lipman et al., §4, Eqs. (10)–(14)

EXERCISE

Conditional Paths

Can we use the endpoint during sampling?

Training uses noise x_0, demonstration x_1 and generation time τ to form the candidate x_τ. A new sampling call has only noise and an observation.

A. Give the network the unknown x_1.

B. Use $x_1 - x_0$ directly at every step.

C. Predict velocity from x_τ, τ and observation.

Lipman et al., §4, Eqs. (10)–(14)

SOLUTION

Conditional Paths

Can we use the endpoint during sampling?

Solution 1 / 1

Separate supervision from inputs

x_1: training target; v_θ: learned velocity.

1. During training x_1 constructs the target $u = x_1 - x_0$.
2. At generation time x_1 is unknown, so u cannot be supplied directly.
3. C is executable. A and B assume the answer is already known.

Lipman et al., §4, Eqs. (10)–(14)

RESULT

Conditional Paths

Can we use the endpoint during sampling?

Training uses noise x_0, demonstration x_1 and generation time τ to form the candidate x_τ. A new sampling call has only noise and an observation.

A. Give the network the unknown x_1.

B. Use $x_1 - x_0$ directly at every step.

C. Predict velocity from x_τ, τ and observation. ✓

Lipman et al., §4, Eqs. (10)–(14)

EXERCISE

Conditional Paths

Which average does the field need?

At a fixed state x and generation time τ, target rates are $+2$ with probability $3/4$ and -2 with probability $1/4$. Which average should the model predict?

A. The unconditional endpoint average.

B. The target mean given this x and τ.

C. Always the most frequent target exactly.

Lipman et al., §3.1, Eq. (8)

SOLUTION

Conditional Paths

Which average does the field need?

Solution 1 / 1

Use the input-conditioned target law

$U \mid x, \tau$ is the stated two-valued random target.

---

1. $\mathbb{E}[U \mid x, \tau] = (3/4)2 + (1/4)(-2) = 1.$
2. B selects the relevant conditional distribution.
3. A discards the input condition. C gives +2, which need not minimize squared error.

Lipman et al., §3.1, Eq. (8)

RESULT

Conditional Paths

Which average does the field need?

At a fixed state x and generation time τ, target rates are $+2$ with probability $3/4$ and -2 with probability $1/4$. Which average should the model predict?

A. The unconditional endpoint average.

B. The target mean given this x and τ. ✓

C. Always the most frequent target exactly.

Lipman et al., §3.1, Eq. (8)

EXERCISE

Learning the Velocity

Transfer: changed conditional weights

At a fixed state x and generation time τ, the target U equals -1 with probability $1/4$ and 3 with probability $3/4$. Minimize conditional squared error.

A Predict 1

B Predict 2

C Predict 3

Lipman et al., §3.2, Eq. (9)

SOLUTION

Learning the Velocity

Transfer: changed conditional weights

Compute the conditional mean

U: scalar rate target; a: prediction.

1. $a^* = (1/4)(-1) + (3/4)3 = 2.$
2. $R(a) = (1/4)(a+1)^2 + (3/4)(a-3)^2.$
3. $R(a) = a^2 - 4a + 7 = (a-2)^2 + 3.$

Lipman et al., §3.2, Eq. (9)

SOLUTION

Learning the Velocity

Transfer: changed conditional weights

Solution 2 / 2

Choose B

$R(2) = 3$, while $R(1) = R(3) = 4$.

1. B minimizes the risk.
2. A uses equal rather than stated weights. C selects the more frequent label rather than the mean.

Lipman et al., §3.2, Eq. (9)

RESULT

Learning the Velocity

Transfer: changed conditional weights

At a fixed state x and generation time τ, the target U equals -1 with probability $1/4$ and 3 with probability $3/4$. Minimize conditional squared error.

A Predict 1

B Predict 2 ✓

C Predict 3

Lipman et al., §3.2, Eq. (9)

EXERCISE

Learning the Velocity

Shapes and loss reduction

Prediction V and target U have $B = 2$ sample rows i and $d = 3$ coordinate columns j. Find the gradient of $L = \frac{1}{Bd}\sum_{i,j}(V_{ij} - U_{ij})^2$.

A. Gradient is 2(V−U)

B. Gradient is (V−U)/3

C. Gradient has shape 3×2

Lipman et al., Theorem 2; course least-squares calculation

SOLUTION

Learning the Velocity

Shapes and loss reduction

Solution 1 / 1

Differentiate the stated mean

V: predictions; U: fixed labels; $Bd = 6$.

1. $\partial L / \partial V_{ij} = 2(V_{ij} - U_{ij})/6$.
2. $\nabla_V L = (V - U)/3 \in \mathbb{R}^{2\times 3}$.
3. B is correct. A omits the reduction; C transposes the gradient shape.

Lipman et al., Theorem 2; course least-squares calculation

RESULT

Learning the Velocity

Shapes and loss reduction

Prediction V and target U have $B = 2$ sample rows i and $d = 3$ coordinate columns j. Find the gradient of $L = \frac{1}{Bd}\sum_{i,j}(V_{ij} - U_{ij})^2$.

A Gradient is 2(V−U)

B Gradient is (V−U)/3 ✓

C Gradient has shape 3×2

Lipman et al., Theorem 2; course least-squares calculation

EXERCISE

Learning the Velocity

Transfer: a negative input

Rate $v_w(x) = wx$ with input $x = -2$, target $u = 3$, and weight $w = 1$. Take one descent step of size $\eta = .1$ on $L = \frac{1}{2}(wx - u)^2$.

A New w is 0

B New w is 2

C New w is 1.5

Lipman et al., Theorem 2; course least-squares calculation

SOLUTION

Learning the Velocity

Transfer: a negative input

Differentiate before updating

$e = wx - u$: residual; w: weight; x, u: fixed input and target.

1. $e = 1(-2) - 3 = -5.$
2. $dL/dw = ex = (-5)(-2) = 10.$
3. $w_{new} = 1 - .1(10) = 0.$

Lipman et al., Theorem 2; course least-squares calculation

SOLUTION

Learning the Velocity

Transfer: a negative input

Solution 2 / 2

Choose A

Old loss $25/2$; new loss $9/2$.

1. A follows gradient descent and decreases this example loss.
2. B uses the wrong update sign. C omits the input factor in the chain rule.

Lipman et al., Theorem 2; course least-squares calculation

RESULT

Learning the Velocity

Transfer: a negative input

Rate $v_w(x) = wx$ with input $x = -2$, target $u = 3$, and weight $w = 1$. Take one descent step of size $\eta = .1$ on $L = \frac{1}{2}(wx - u)^2$.

- **A** New w is 0 ✓
- **B** New w is 2
- **C** New w is 1.5

Lipman et al., Theorem 2; course least-squares calculation

EXERCISE

Sampling the Learned Flow

What remains fixed while sampling?

Candidate x_τ follows $dx_\tau/d\tau = v_\theta(x_\tau, \tau, c)$. Draw x_0 once. τ: generation time; θ: field weights; c: fixed observation.

A. Optimize θ at every flow step.

B. Hold θ and c fixed; update x and τ.

C. Replace x with fresh noise every step.

Lecture 7, p22 and pp33–34; Lipman et al., §2

SOLUTION

Sampling the Learned Flow

What remains fixed while sampling?

Solution 1 / 1

Read the differential equation

θ: learned parameters; c: condition; x_τ: generated state.

1. The ODE evolves x with generation time τ under one fixed field.
2. B follows the stated ODE. A adds test-time training; C defines a different stochastic process.
3. The initial draw supplies randomness even with deterministic integration.

Lecture 7, p22 and pp33–34; Lipman et al., §2

RESULT

Sampling the Learned Flow

What remains fixed while sampling?

Candidate x_τ follows $dx_\tau/d\tau = v_\theta(x_\tau, \tau, c)$. Draw x_0 once. τ: generation time; θ: field weights; c: fixed observation.

A. Optimize θ at every flow step.

B. Hold θ and c fixed; update x and τ. ✓

C. Replace x with fresh noise every step.

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Sampling the Learned Flow

Transfer: a time-dependent rate

State x starts at $x_0 = 1$. The rate is $v(x, \tau) = 2\tau$. Use two forward Euler steps of size $h = 1/2$ from time $\tau = 0$ to 1.

A 1.5

B 2

C 2.5

Lecture 7, p22 and pp33–34; Lipman et al., §2

SOLUTION

Sampling the Learned Flow

Transfer: a time-dependent rate

Evaluate at the beginning of each step

x_j: Euler state; $\tau_j = j/2$.

1. $v(x_0, 0) = 0$, so $x_1 = 1 + (1/2)0 = 1$.
2. $v(x_1, 1/2) = 1$, so $x_2 = 1 + (1/2)1 = 1.5$.

Lecture 7, p22 and pp33–34; Lipman et al., §2

SOLUTION

Sampling the Learned Flow

Transfer: a time-dependent rate

Choose A and compare the exact integral

$x(1) = 1 + \int_0^1 2\tau \, d\tau = 2.$

1. A is the forward Euler approximation.
2. B is exact here, not two-step forward Euler. C uses right-end rates.

Lecture 7, p22 and pp33–34; Lipman et al., §2

RESULT

Sampling the Learned Flow

Transfer: a time-dependent rate

State x starts at $x_0 = 1$. The rate is $v(x, \tau) = 2\tau$. Use two forward Euler steps of size $h = 1/2$ from time $\tau = 0$ to 1.

- **A** 1.5 ✓
- **B** 2
- **C** 2.5

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Sampling the Learned Flow

Transfer: a fixed evaluation budget

Budget: 8 calls to learned rate v_θ. Euler uses 1 call/step; explicit midpoint uses 2. Both integrate $[0,1]$. Compare step sizes h.

A Euler h=1/8; midpoint h=1/4

B Both h=1/8

C Midpoint always gives the smaller error

Lecture 7, p22 and pp33–34; Lipman et al., §2

SOLUTION

Sampling the Learned Flow

Transfer: a fixed evaluation budget

Solution 1 / 1

Count steps from calls per step

N: accepted steps; $h = 1/N$.

1. Euler: $N = 8/1 = 8$, so $h = 1/8$.
2. Midpoint: $N = 8/2 = 4$, so $h = 1/4$.
3. A uses the full budget. B needs 16 midpoint calls; C needs assumptions and an error comparison.

Lecture 7, p22 and pp33–34; Lipman et al., §2

RESULT

Sampling the Learned Flow

Transfer: a fixed evaluation budget

Budget: 8 calls to learned rate v_θ. Euler uses 1 call/step; explicit midpoint uses 2. Both integrate $[0,1]$. Compare step sizes h.

- **A** Euler h=1/8; midpoint h=1/4 ✓
- **B** Both h=1/8
- **C** Midpoint always gives the smaller error

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Sampling the Learned Flow

Reversing the generation clock

Course candidate x_τ has rate $v(x_\tau, \tau)$. Reverse the clock by $s = 1 - \tau$ and set $y_s = x_{1-s}$. Find the new rate dy_s/ds.

A $v(y_s, 1-s)$

B $-v(y_s, 1-s)$

C $-v(y_s, s)$ always

Lecture 7, p22 and pp33–34; Lipman et al., §2

SOLUTION

Sampling the Learned Flow

Reversing the generation clock

Solution 1 / 1

Apply the chain rule

s: reversed clock; $y_s = x_{1-s}$.

1. $dy_s/ds = (dx_\tau/d\tau)(d\tau/ds)$.
2. $d\tau/ds = -1$, hence $dy_s/ds = -v(y_s, 1-s)$.
3. B changes both the sign and time argument. A misses the sign; C generally uses the wrong time.

Lecture 7, p22 and pp33–34; Lipman et al., §2

RESULT

Sampling the Learned Flow

Reversing the generation clock

Course candidate x_τ has rate $v(x_\tau, \tau)$. Reverse the clock by $s = 1 - \tau$ and set $y_s = x_{1-s}$. Find the new rate dy_s/ds.

A $v(y_s, 1-s)$

B $-v(y_s, 1-s)$ ✓

C $-v(y_s, s)$ always

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Pairing and Optimal Transport

Transfer: compute the assignment cost

Sources $0, 2$; targets $1, 4$; each has mass $1/2$. Compare aligned $(0 \to 1, 2 \to 4)$ and crossed pairs.

A. Aligned cost 2.5; crossed 8.5

B. Aligned cost 5; crossed 17

C. Both cost 2.5

Tong et al., §§2.2, 3.2.3

SOLUTION

Pairing and Optimal Transport

Transfer: compute the assignment cost

Square each displacement, then average

C is expected squared cost, not a sum.

1. $C_{aligned} = [(1-0)^2 + (4-2)^2]/2 = (1+4)/2 = 2.5.$
2. $C_{crossed} = [(4-0)^2 + (1-2)^2]/2 = (16+1)/2 = 8.5.$

Tong et al., §§2.2, 3.2.3

SOLUTION

Pairing and Optimal Transport

Transfer: compute the assignment cost

Solution 2 / 2

Choose A

Both assignments preserve one-half mass at each endpoint.

1. A includes the probability weights.
2. B reports sums, not expected costs. C assumes fixed marginals determine the joint pairing.

Tong et al., §§2.2, 3.2.3

RESULT

Pairing and Optimal Transport

Transfer: compute the assignment cost

Sources $0, 2$; targets $1, 4$; each has mass $1/2$. Compare aligned $(0 \to 1, 2 \to 4)$ and crossed pairs.

- **A** Aligned cost 2.5; crossed 8.5 ✓
- **B** Aligned cost 5; crossed 17
- **C** Both cost 2.5

Tong et al., §§2.2, 3.2.3

EXERCISE

Pairing and Optimal Transport

Are learned trajectories necessarily straight?

Paired training paths are straight. At state x and time τ, $v^*(x,\tau) = \mathbb{E}[U|x,\tau]$ averages target rates U. Are generated paths straight?

A Yes, every generated path is one training segment.

B No; the conditional average can vary along a path.

C Every learned path must cross another path.

Lipman et al., §3.1, Eq. (8)

SOLUTION

Pairing and Optimal Transport

Are learned trajectories necessarily straight?

Solution 1 / 1

Track the field along the generated state

U: paired velocity label; v^*: conditional mean field.

1. As x and τ change, the compatible hidden endpoint pairs can change.
2. Their conditional average can therefore change direction along a generated trajectory.
3. B is correct. A confuses paired paths with the learned flow; C contradicts uniqueness for a regular ODE.

Lipman et al., §3.1, Eq. (8)

RESULT

Pairing and Optimal Transport

Are learned trajectories necessarily straight?

Paired training paths are straight. At state x and time τ, $v^*(x,\tau) = \mathbb{E}[U|x,\tau]$ averages target rates U. Are generated paths straight?

A. Yes, every generated path is one training segment.

B. No; the conditional average can vary along a path. ✓

C. Every learned path must cross another path.

Lipman et al., §3.1, Eq. (8)

EXERCISE

Pairing and Optimal Transport

What does a conditional path establish?

Each conditional path moves Gaussian noise toward a fixed demonstration. Noise and demonstrations may be paired independently. What does this establish?

A. This alone proves globally optimal endpoint pairing.

B. A simple conditional path and global OT are different claims.

C. Every coupling has equal expected displacement cost.

Tong et al., §3.2.1, after Eq. (13)

SOLUTION

Pairing and Optimal Transport

What does a conditional path establish?

Solution 1 / 1

Identify what was optimized

y: fixed endpoint in a conditional path; π: joint endpoint pairing.

1. A simple path conditional on y does not choose the best joint pairing π.
2. Global quadratic OT additionally minimizes expected cost over couplings with fixed marginals.
3. B is correct. A skips that optimization; C fails in the 1-versus-9 cost example.

Tong et al., §3.2.1, after Eq. (13)

RESULT

Pairing and Optimal Transport

What does a conditional path establish?

Each conditional path moves Gaussian noise toward a fixed demonstration. Noise and demonstrations may be paired independently. What does this establish?

A. This alone proves globally optimal endpoint pairing.

B. A simple conditional path and global OT are different claims. ✓

C. Every coupling has equal expected displacement cost.

Tong et al., §3.2.1, after Eq. (13)

EXERCISE

Pairing and Optimal Transport

Re-pair examples with different conditions

Condition c says “move left”. Its data actions are negative. A batch also contains “move right” examples.

A. Re-pair all targets freely and retain old c.

B. Keep targets compatible with each condition.

C. Remove c from training and still claim conditioning.

Tong et al., §§3.1–3.2; course conditional-pairing example

SOLUTION

Pairing and Optimal Transport

Re-pair examples with different conditions

Solution 1 / 1

Preserve the conditional target law

$p_{data}(x_1 \mid c)$: action distribution for a fixed condition.

1. The conditional generator must learn endpoint actions paired with their observation c.
2. B retains the intended conditional law.
3. A can relabel a right action as left. C trains an unconditional model instead.

Tong et al., §§3.1–3.2; course conditional-pairing example

RESULT

Pairing and Optimal Transport

Re-pair examples with different conditions

Condition c says “move left”. Its data actions are negative. A batch also contains “move right” examples.

A. Re-pair all targets freely and retain old c.

B. Keep targets compatible with each condition. ✓

C. Remove c from training and still claim conditioning.

Tong et al., §§3.1–3.2; course conditional-pairing example

EXERCISE

The Bridge to Diffusion

Worked bridge: a known Gaussian path

Straight path $X_\tau = (1-\tau)\varepsilon + \tau Y$, fixed data $Y = 2$, standard Gaussian noise ε. At $\tau = 1/2$, $X_\tau \sim \mathcal{N}(1, 1/4)$ (variance $1/4$). Find score s and rate v at $x = 0$.

A s=4, v=4

B s=2, v=2

C s=−4, v=−4

Lipman et al., §4 and Appendix D; course score derivation

SOLUTION

The Bridge to Diffusion

Worked bridge: a known Gaussian path

Differentiate the Gaussian log density

$\mu = 1$, $\sigma^2 = 1/4$; $s(x) = -(x - \mu)/\sigma^2$.

1. $s(0) = -(0 - 1)/(1/4) = 4.$
2. $v(0, 1/2) = 0/(1/2) + [(1/2)/(1/2)]4 = 4.$

Lipman et al., §4 and Appendix D; course score derivation

SOLUTION

The Bridge to Diffusion

Worked bridge: a known Gaussian path

Check by the endpoint-conditioned field

$y = 2$, $x = 0$, $\tau = 1/2$.

1. $u = (y - x)/(1 - \tau) = 2/(1/2) = 4$.
2. A agrees with both derivations. B uses standard deviation as variance; C reverses the Gaussian score sign.

Lipman et al., §4 and Appendix D; course score derivation

RESULT

The Bridge to Diffusion

Worked bridge: a known Gaussian path

Straight path $X_\tau = (1-\tau)\varepsilon + \tau Y$, fixed data $Y = 2$, standard Gaussian noise ε. At $\tau = 1/2$, $X_\tau \sim \mathcal{N}(1, 1/4)$ (variance $1/4$). Find score s and rate v at $x = 0$.

- **A** s=4, v=4 ✓
- **B** s=2, v=2
- **C** s=−4, v=−4

Lipman et al., §4 and Appendix D; course score derivation

EXERCISE

The Bridge to Diffusion

Transfer: reverse the clock consistently

At course time $\tau = 1/2$ the rate is $+4$. Code time is $s = 1 - \tau$. For an Euler step $\Delta s = -.1$, find the code rate and state increment.

A Code velocity -4, state increment $+.4$

B Code velocity $+4$, state increment $-.4$

C Code velocity -4, state increment $-.4$

openpi 5546575: compute_loss / sample_actions

SOLUTION

The Bridge to Diffusion

Transfer: reverse the clock consistently

Solution 1 / 1

Transform rate and increment together

$w(x, s) = -v(x, 1 - s)$; $\Delta s = -.1$.

1. $w = -4$.
2. $\Delta x \approx \Delta s\, w = (-.1)(-4) = +.4$.
3. A preserves the same forward generation direction. B misses the field sign; C multiplies signs incorrectly.

openpi 5546575: compute_loss / sample_actions

RESULT

The Bridge to Diffusion

Transfer: reverse the clock consistently

At course time $\tau = 1/2$ the rate is $+4$. Code time is $s = 1 - \tau$. For an Euler step $\Delta s = -.1$, find the code rate and state increment.

- **A** Code velocity -4, state increment $+.4$ ✓
- **B** Code velocity $+4$, state increment $-.4$
- **C** Code velocity -4, state increment $-.4$

openpi 5546575: compute_loss / sample_actions

EXERCISE

Vision-Language-Action Models

A code step toward the data

Target $A = 2$, noise $\varepsilon = -1$. At code time $s = 1$, candidate $z = -1$. Find target rate u and the candidate after a step $\Delta s = -1/4$.

A u=−3, new z=−.25

B u=+3, new z=−1.75

C u=−3, new z=−1.75

openpi 5546575: Pi0.compute_loss / sample_actions

SOLUTION

Vision-Language-Action Models

A code step toward the data

Solution 1 / 2

Follow the code convention

$u = \varepsilon - A$; $z_{new} = z + \Delta s\, u$.

1. $u = -1 - 2 = -3.$
2. $z_{new} = -1 + (-1/4)(-3) = -1 + 3/4 = -1/4.$

openpi 5546575: Pi0.compute_loss / sample_actions

SOLUTION

Vision-Language-Action Models

A code step toward the data

Solution 2 / 2

Check the interpolation at the new time

$s_{new} = 3/4$; $Z_s = (1-s)A + s\varepsilon$.

1. $Z_{3/4} = (1/4)2 + (3/4)(-1) = -1/4$.
2. A agrees. B reverses the target; C reverses the increment sign.

openpi 5546575: Pi0.compute_loss / sample_actions

RESULT

Vision-Language-Action Models

A code step toward the data

Target $A = 2$, noise $\varepsilon = -1$. At code time $s = 1$, candidate $z = -1$. Find target rate u and the candidate after a step $\Delta s = -1/4$.

- **A** u=−3, new z=−.25 ✓
- **B** u=+3, new z=−1.75
- **C** u=−3, new z=−1.75

[openpi 5546575: Pi0.compute_loss / sample_actions]

EXERCISE

Vision-Language-Action Models

Generation, execution, and replanning

Generate $H = 6$ actions. Execute the first $r = 2$, observe again, then generate a new chunk.

A. Six internal steps imply six robot actions.

B. Physical time advances by two executed actions.

C. The remaining four must always be executed.

π_0, §IV; Appendix A-B and A-D

SOLUTION

Vision-Language-Action Models

Generation, execution, and replanning

Solution 1 / 1

Follow the physical loop

H: generated horizon; r: executed prefix length; $1 \leq r \leq H$.

1. The internal solver first produces a six-action candidate.
2. Executing two entries advances the robot by two action periods.
3. B is correct. A confuses clocks. C rules out the stated replanning procedure.

π_0, §IV; Appendix A-B and A-D

RESULT

Vision-Language-Action Models

Generation, execution, and replanning

Generate $H = 6$ actions. Execute the first $r = 2$, observe again, then generate a new chunk.

A Six internal steps imply six robot actions.

B Physical time advances by two executed actions. ✓

C The remaining four must always be executed.

π_0, §IV; Appendix A-B and A-D

EXERCISE

Vision-Language-Action Models

What follows from a richer training mixture?

A vision-language-action model receives additional image-language and robot examples. Its low-level action model still uses flow matching.

A. The ODE becomes physical robot dynamics.

B. Representation learning changes; the flow objective can remain.

C. Generalization is guaranteed for every new home.

Π0.5, §§IV-C–D, V-C

SOLUTION

Vision-Language-Action Models

What follows from a richer training mixture?

Solution 1 / 1

Separate the mathematical objective from evidence

VLA: vision-language-action model; ODE: numerical action-generation flow.

1. Additional data changes the learned representation and parameters.
2. It does not redefine generation time as physical time or prove every new task succeeds.
3. B follows. A confuses the two systems; C overstates empirical evidence.

Пo.5, §§IV-C–D, V-C

RESULT

Vision-Language-Action Models

What follows from a richer training mixture?

A vision-language-action model receives additional image-language and robot examples. Its low-level action model still uses flow matching.

A. The ODE becomes physical robot dynamics.

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C. Generalization is guaranteed for every new home.

Π0.5, §§IV-C–D, V-C

EXERCISE

From Data to an Executed Action

Transfer: the observation changes mid-execution

After one executed action, a new image shows an obstacle. Old condition c described a clear route.

A. More ODE steps under old c incorporate the new image.

B. Update the condition and replan under the policy.

C. Low training MSE guarantees the old chunk is safe.

π_0, §IV and Appendix A-D; original course example

SOLUTION

From Data to an Executed Action

Transfer: the observation changes mid-execution

Solution 1 / 1

Identify the changed input

c_{old} and c_{new} represent different observations.

1. Continuing integration under c_old keeps using the old information.
2. B passes the new condition into a new policy computation.
3. A confuses numerical accuracy with updated observations; C confuses regression fit with a safety guarantee.

π_0, §IV and Appendix A-D; original course example

RESULT

From Data to an Executed Action

Transfer: the observation changes mid-execution

After one executed action, a new image shows an obstacle. Old condition c described a clear route.

A More ODE steps under old c incorporate the new image.

B Update the condition and replan under the policy. ✓

C Low training MSE guarantees the old chunk is safe.

Π_0, §IV and Appendix A-D; original course example